

IT1223(P) - Database Management Systems (P)

Department of Physical Science

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Revision Worksheet - 02

Answers

Create a database named CompanyDB and use for the below operations:

1. Create a MySQL tables named "Employee", "Department", "Project", "Works_in", "Manager" and

"Works_on" with appropriate fields to store company information.

Create database

CREATE DATABASE CompanyDB;

Use the database

USE CompanyDB;

```
MariaDB [(none)]> USE CompanyDB;  
Database changed
```

Create the Employee table

CREATE TABLE Employee (

Id INT PRIMARY KEY,

Name VARCHAR(100),

Gender VARCHAR(10),

Address VARCHAR(255),

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Dob DATE,

Doj DATE

);

-- Create the Department table

CREATE TABLE Department (

D_No INT PRIMARY KEY,

Name VARCHAR(100),

Location VARCHAR(100)

);

-- Create the Project table

CREATE TABLE Project (

P_No INT PRIMARY KEY,

Name VARCHAR(100),

Location VARCHAR(100)

);

-- Create the Works_in relationship table

CREATE TABLE Works_in (

Employee_Id INT,

Department_No INT,

Since DATE,

PRIMARY KEY (Employee_Id, Department_No),

FOREIGN KEY (Employee_Id) REFERENCES Employee(Id),

FOREIGN KEY (Department_No) REFERENCES Department(D_No)

);

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-- Create the Manager relationship table

```
CREATE TABLE Manager (  
    Department_No INT,  
    Employee_Id INT  
);
```

-- Create the Works_on relationship table

```
CREATE TABLE Works_on (  
    Employee_Id INT,  
    Project_No INT,  
    Hours INT,  
    PRIMARY KEY (Employee_Id, Project_No),  
    FOREIGN KEY (Employee_Id) REFERENCES Employee(Id),  
    FOREIGN KEY (Project_No) REFERENCES Project(P_No)  
);
```

```
MariaDB [(none)]> USE CompanyDB;
Database changed
MariaDB [CompanyDB]> CREATE TABLE Employee (
    ->     Id INT PRIMARY KEY,
    ->     Name VARCHAR(100),
    ->     Gender VARCHAR(10),
    ->     Address VARCHAR(255),
    ->     Dob DATE,
    ->     Doj DATE
    -> );
Query OK, 0 rows affected (0.017 sec)

MariaDB [CompanyDB]> CREATE TABLE Department (
    ->     D_No INT PRIMARY KEY,
    ->     Name VARCHAR(100),
    ->     Location VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.030 sec)

MariaDB [CompanyDB]> CREATE TABLE Project (
    ->     P_No INT PRIMARY KEY,
    ->     Name VARCHAR(100),
    ->     Location VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.023 sec)

MariaDB [CompanyDB]> CREATE TABLE Works_in (
    ->     Employee_Id INT,
    ->     Department_No INT,
    ->     Since DATE,
    ->     PRIMARY KEY (Employee_Id, Department_No),
    ->     FOREIGN KEY (Employee_Id) REFERENCES Employee(Id),
    ->     FOREIGN KEY (Department_No) REFERENCES Department(D_No)
    -> );
Query OK, 0 rows affected (0.031 sec)

MariaDB [CompanyDB]> LOAD DATA INFILE 'C:/Users/User/Desktop/manager.txt'
    -> INTO TABLE Manager
    -> FIELDS TERMINATED BY ','
    -> LINES TERMINATED BY '\n'
    -> IGNORE 1 ROWS (Department_No, Employee_Id)
    -> ;
Query OK, 4 rows affected, 3 warnings (0.009 sec)

MariaDB [CompanyDB]> CREATE TABLE Works_on (
    ->     Employee_Id INT,
    ->     Project_No INT,
    ->     Hours INT,
    ->     PRIMARY KEY (Employee_Id, Project_No),
    ->     FOREIGN KEY (Employee_Id) REFERENCES Employee(Id),
    ->     FOREIGN KEY (Project_No) REFERENCES Project(P_No)
    -> );
Query OK, 0 rows affected (0.031 sec)
```

2. Write a query to view the structure of the "Project" table in MySQL, specifying the data types and

constraints for each field.

DESCRIBE Project;

```
MariaDB [CompanyDB]> DESCRIBE Project;
```

Field	Type	Null	Key	Default	Extra
P_No	int(11)	NO	PRI	NULL	
Name	varchar(100)	YES		NULL	
Location	varchar(100)	YES		NULL	

```
3 rows in set (0.008 sec)

MariaDB [CompanyDB]> |
```

3. Write a SQL query to insert the values into table "Employee", "Department", "Project", "Works_In", "Manager" and "Works_On" from the local file which contains comma separated values (Company.txt).

```
LOAD DATA INFILE 'C:/Users/User/Desktop/EMPLOYEE.txt'
INTO TABLE Employee
FIELDS TERMINATED BY ','
LINES TERMINATED BY '\n'
;
```

```
LOAD DATA INFILE 'C:/Users/User/Desktop/DEPARTMENT.txt'
INTO TABLE Department
FIELDS TERMINATED BY ','
LINES TERMINATED BY '\n'
;
```

```
LOAD DATA INFILE 'C:/Users/User/Desktop/PROJECT.txt'
```

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INTO TABLE Project

FIELDS TERMINATED BY ','

LINES TERMINATED BY '\n'

;

LOAD DATA INFILE 'C:/Users/User/Desktop/works_in.txt'

INTO TABLE Works_in

FIELDS TERMINATED BY ','

LINES TERMINATED BY '\n'

;

LOAD DATA INFILE 'C:/Users/User/Desktop/works_on.txt'

INTO TABLE Works_on

FIELDS TERMINATED BY ','

LINES TERMINATED BY '\n'

;

```
MariaDB [CompanyDB]> LOAD DATA INFILE 'C:/Users/User/Desktop/EMPLOYEE.txt'
-> INTO TABLE Employee
-> FIELDS TERMINATED BY ','
-> LINES TERMINATED BY '\n'
-> ;
Query OK, 5 rows affected, 20 warnings (0.011 sec)
Records: 5 Deleted: 0 Skipped: 0 Warnings: 20

MariaDB [CompanyDB]> LOAD DATA INFILE 'C:/Users/User/Desktop/DEPARTMENT.txt'
-> INTO TABLE Department
-> FIELDS TERMINATED BY ','
-> LINES TERMINATED BY '\n'
-> ;
Query OK, 5 rows affected, 4 warnings (0.008 sec)
Records: 5 Deleted: 0 Skipped: 0 Warnings: 4

MariaDB [CompanyDB]> LOAD DATA INFILE 'C:/Users/User/Desktop/PROJECT.txt'
-> INTO TABLE Project
-> FIELDS TERMINATED BY ','
-> LINES TERMINATED BY '\n'
-> ;
Query OK, 5 rows affected, 4 warnings (0.009 sec)
Records: 5 Deleted: 0 Skipped: 0 Warnings: 4

MariaDB [CompanyDB]> LOAD DATA INFILE 'C:/Users/User/Desktop/works_in.txt'
-> INTO TABLE Works_in
-> FIELDS TERMINATED BY ','
-> LINES TERMINATED BY '\n'
-> ;
Query OK, 5 rows affected, 9 warnings (0.009 sec)
Records: 5 Deleted: 0 Skipped: 0 Warnings: 9

MariaDB [CompanyDB]> LOAD DATA INFILE 'C:/Users/User/Desktop/manager.txt'
-> INTO TABLE Manager
-> FIELDS TERMINATED BY ','
-> LINES TERMINATED BY '\n'
-> IGNORE 1 ROWS (Department_No, Employee_Id)
-> ;
Query OK, 4 rows affected, 3 warnings (0.009 sec)
Records: 4 Deleted: 0 Skipped: 0 Warnings: 3
```

4. Retrieve the male names of all employees.

SELECT Name FROM Employee WHERE Gender = 'Male';

```
MariaDB [CompanyDB]> SELECT Name FROM Employee WHERE Gender = 'Male';
Empty set (0.001 sec)
```

5. Retrieve the names of all employees, sorted by their date of joining.

SELECT Name FROM Employee ORDER BY Doj;

```
MariaDB [CompanyDB]> SELECT Name FROM Employee ORDER BY Doj;
+-----+
| Name |
+-----+
| 'John Doe' |
| 'Jane Smith' |
| 'Alice Johnson' |
| 'Bob Brown' |
| 'Charlie Davis' |
+-----+
5 rows in set (0.001 sec)

MariaDB [CompanyDB]> |
```

6. Write a SQL query to find all female employees who joined after 2019.

```
SELECT Name FROM Employee WHERE Gender = 'Female' AND Doj > '2019-01-01';
```

```
MariaDB [CompanyDB]> SELECT Name FROM Employee WHERE Gender = 'Female' AND Doj > '2019-01-01';
Empty set (0.001 sec)
```

7. Find the oldest and youngest employees based on date of birth.

```
SELECT Name, Dob FROM Employee ORDER BY Dob ASC LIMIT 1; -- Oldest employee
```

```
SELECT Name, Dob FROM Employee ORDER BY Dob DESC LIMIT 1; -- Youngest employee;
```

```
MariaDB [CompanyDB]> SELECT Name, Dob FROM Employee ORDER BY Dob DESC LIMIT 1; -- Youngest employee;
+-----+-----+
| Name | Dob |
+-----+-----+
| 'Charlie Davis' | 0000-00-00 |
+-----+-----+
1 row in set (0.000 sec)
```

8. Calculate the total, average, minimum, and maximum hours worked by employees on projects.

```
SELECT
```

```
    SUM(Hours) AS Total_Hours,
```

```
    AVG(Hours) AS Average_Hours,
```

```
    MIN(Hours) AS Minimum_Hours,
```

```
    MAX(Hours) AS Maximum_Hours
```

```
FROM Works_on;
```



```
MariaDB [CompanyDB]> SELECT
->     SUM(Hours) AS Total_Hours,
->     AVG(Hours) AS Average_Hours,
->     MIN(Hours) AS Minimum_Hours,
->     MAX(Hours) AS Maximum_Hours
-> FROM Works_on;
+-----+-----+-----+-----+
| Total_Hours | Average_Hours | Minimum_Hours | Maximum_Hours |
+-----+-----+-----+-----+
|          150 |          30.0000 |             20 |             40 |
+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

9. Retrieve the first 3 projects.

SELECT * FROM Project LIMIT 3;

```
MariaDB [CompanyDB]> SELECT * FROM Project LIMIT 3;
+-----+-----+-----+
| P_No | Name          | Location    |
+-----+-----+-----+
| 1001 | 'Project Alpha' | 'New York'  |
| 1002 | 'Project Beta'  | 'California' |
| 1003 | 'Project Gamma' | 'Texas'     |
+-----+-----+-----+
3 rows in set (0.001 sec)
```

10. Find employees who live in either 'NY' or 'CA' and whose date of birth is between '1980-0101' and

'1990-12-31'.

SELECT * FROM Employee

WHERE (Address LIKE '%NY%' OR Address LIKE '%CA%')

AND Dob BETWEEN '1980-01-01' AND '1990-12-31';

```
MariaDB [CompanyDB]> SELECT * FROM Employee
-> WHERE (Address LIKE '%NY%' OR Address LIKE '%CA%')
-> AND Dob BETWEEN '1980-01-01' AND '1990-12-31';
Empty set (0.001 sec)
```

11. Find the number of employees in each department.

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SELECT Department_No, COUNT(Employee_Id) AS NumberOfEmployees

FROM Works_in

GROUP BY Department_No;

```
MariaDB [CompanyDB]> SELECT Department_No, COUNT(Employee_Id) AS NumberOfEmployees
-> FROM Works_in
-> GROUP BY Department_No;
+-----+-----+
| Department_No | NumberOfEmployees |
+-----+-----+
| 101           | 1                 |
| 102           | 1                 |
| 103           | 1                 |
| 104           | 1                 |
| 105           | 1                 |
+-----+-----+
5 rows in set (0.001 sec)
```

12. Find all employees whose name starts with 'A'.

SELECT * FROM Employee WHERE Name LIKE 'A%';

```
MariaDB [CompanyDB]> SELECT * FROM Employee WHERE Name LIKE 'A%';
Empty set (0.001 sec)
```

13. Retrieve all employee names along with their department names.

SELECT e.Name AS EmployeeName, d.Name AS DepartmentName

FROM Employee e

JOIN Works_in w ON e.Id = w.Employee_Id

JOIN Department d ON w.Department_No = d.D_No;

```
MariaDB [CompanyDB]> SELECT e.Name AS EmployeeName, d.Name AS DepartmentName
-> FROM Employee e
-> JOIN Works_in w ON e.Id = w.Employee_Id
-> JOIN Department d ON w.Department_No = d.D_No;
+-----+-----+
| EmployeeName | DepartmentName |
+-----+-----+
| 'John Doe'   | 'HR'           |
| 'Jane Smith' | 'IT'           |
| 'Alice Johnson' | 'Finance'      |
| 'Bob Brown'  | 'Marketing'     |
| 'Charlie Davis' | 'Operations'    |
+-----+-----+
5 rows in set (0.002 sec)
```

14. Retrieve all projects and the employees working on them, if any.

SELECT p.Name AS ProjectName, e.Name AS EmployeeName

FROM Project p

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LEFT JOIN Works_on w ON p.P_No = w.Project_No

LEFT JOIN Employee e ON w.Employee_Id = e.Id;

```
MariaDB [CompanyDB]> SELECT p.Name AS ProjectName, e.Name AS EmployeeName
-> FROM Project p
-> LEFT JOIN Works_on w ON p.P_No = w.Project_No
-> LEFT JOIN Employee e ON w.Employee_Id = e.Id;
```

ProjectName	EmployeeName
'Project Alpha'	'John Doe'
'Project Beta'	'Jane Smith'
'Project Gamma'	'Alice Johnson'
'Project Delta'	'Bob Brown'
'Project Epsilon'	'Charlie Davis'

5 rows in set (0.001 sec)

15. Retrieve all employees and their dependents, if any.

SELECT e.Name AS EmployeeName, d.Name AS DependentName

FROM Employee e

LEFT JOIN Dependents d ON e.Id = d.Employee_Id;

16. Generate a list of all possible pairs of employees and departments.

SELECT e.Name AS EmployeeName, d.Name AS DepartmentName

FROM Employee e

CROSS JOIN Department d;

```
MariaDB [CompanyDB]> SELECT e.Name AS EmployeeName, d.Name AS DepartmentName
-> FROM Employee e
-> CROSS JOIN Department d;
```

EmployeeName	DepartmentName
'John Doe'	'HR'
'Jane Smith'	'HR'
'Alice Johnson'	'HR'
'Bob Brown'	'HR'
'Charlie Davis'	'HR'
'John Doe'	'IT'
'Jane Smith'	'IT'
'Alice Johnson'	'IT'
'Bob Brown'	'IT'
'Charlie Davis'	'IT'
'John Doe'	'Finance'
'Jane Smith'	'Finance'
'Alice Johnson'	'Finance'
'Bob Brown'	'Finance'
'Charlie Davis'	'Finance'
'John Doe'	'Marketing'
'Jane Smith'	'Marketing'
'Alice Johnson'	'Marketing'
'Bob Brown'	'Marketing'
'Charlie Davis'	'Marketing'
'John Doe'	'Operations'
'Jane Smith'	'Operations'
'Alice Johnson'	'Operations'
'Bob Brown'	'Operations'
'Charlie Davis'	'Operations'

```
25 rows in set (0.000 sec)
```

17. Find pairs of employees who work in the same department.

```
SELECT e1.Name AS Employee1, e2.Name AS Employee2, w.Department_No
FROM Works_in w
JOIN Employee e1 ON w.Employee_Id = e1.Id JOIN
Employee e2 ON w.Employee_Id = e2.Id
WHERE e1.Id < e2.Id;
```

```
MariaDB [CompanyDB]> SELECT e1.Name AS Employee1, e2.Name AS Employee2, w.Department_No
-> FROM Works_in w
-> JOIN Employee e1 ON w.Employee_Id = e1.Id
-> JOIN Employee e2 ON w.Employee_Id = e2.Id
-> WHERE e1.Id < e2.Id;
Empty set (0.001 sec)
```

```
MariaDB [CompanyDB]> |
```

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18. Write a query to back up the database and restore it back again. `mysqldump -u [username] -p CompanyDB > CompanyDB_backup.sql` `mysql -u [username] -p CompanyDB < CompanyDB_backup.sql`